

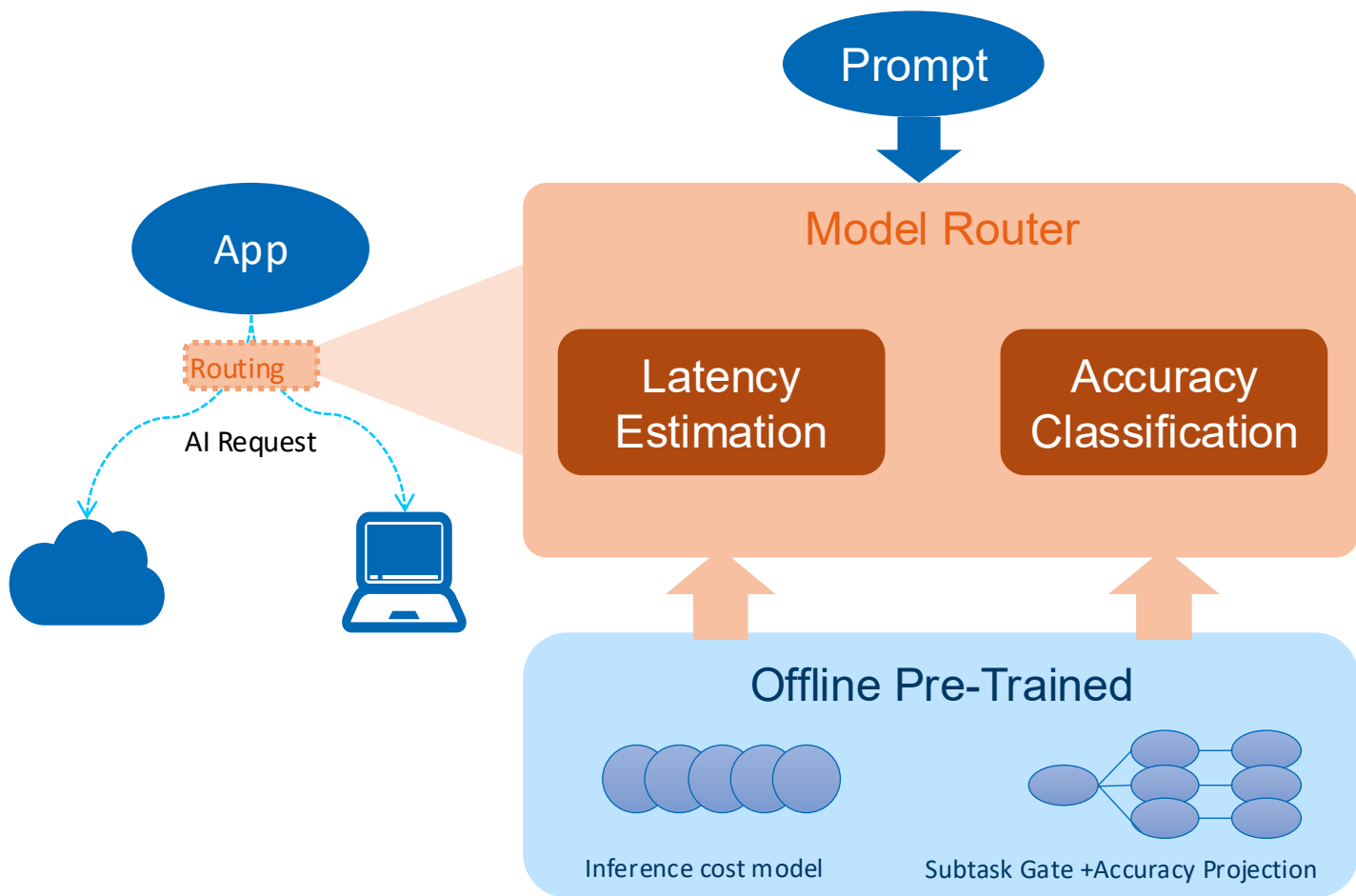
DAOP Update

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DAOP Exploration for Hybrid LLM / Agent scenarios

Dynamic AI Offloading Protocol



Dynamically offload to device / fallback to cloud

Routing based on (latency, accuracy)

- Latency = Estimate(Prompt, Model)
- Accuracy = $P(\text{Correct} \mid \text{Prompt}, \text{Model})$
- Others (cost, privacy, ...) in future

PoC WIP – pre-trained offline models for routing

- Latency based on static cost models with microbenchmarks / profiling data on operators
- Accuracy based on matrix factorization algorithms w/ selected LLM benchmarks – revised on [RouteLLM](#)

Deployment Considerations

- Proposing W3C standards in WebML WG – QoS, Events
- Routing Layer on top of Built-in AI of browser / Web runtime

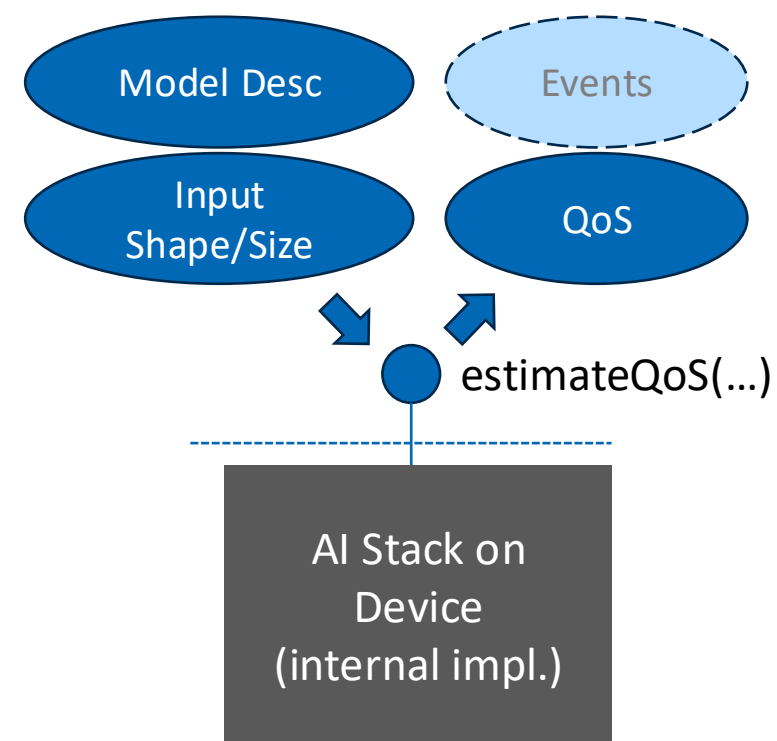
Latency Estimation

Estimation Model

- Static Cost Model on Raw / Fused Compute Graph
- Piecewise Log-space Interpolation in Parameter Space for key Ops, e.g.
 - (M, K, N, dtype) for MUL_MAT
 - (seqQ, seqKV, Q-heads, KV-heads) for FLASH_ATTN

Data Sources

- COLD: One-time microbenchmarks (~90s, ~1000 pts)
- WARM: Profiling data from daily inference [future]



Initial Experiments about Accuracy based offloading

Two-Level Conservative Routing Decision Flow

